carbon dioxide diffuses out. The skin is kept moist and slimy due to secretion of mucus which also prevents it from drying.

13.1.8 Circulatory System

In leech, circulation is brought about by **haemocoelic system**. There are no true blood vessels. The blood vessels are replaced by channels called **haemocoelic channels** or **canals** filled with blood like fluid. The coelomic fluid contains haemoglobin.

There are four longitudinal channels. One channel lies above (dorsal) the alimentary canal, one below (ventral) the alimentary canal. The other two channels lie on either (lateral) side of the alimentary canal which serve as heart and have inner valves. All the four channels are connected together posteriorly in the 26th segment.

13.1.9 Nervous System

The central nervous system of leech consists of a nerve ring and a paired ventral nerve cord. The nerve ring surrounds the pharynx and is formed of **suprapharyngeal ganglion** (brain), **circumpharyngeal connective** and **subpharyngeal ganglion**. The subpharyngeal ganglion lies below the pharynx and is formed by the fusion of four pairs of ganglia.

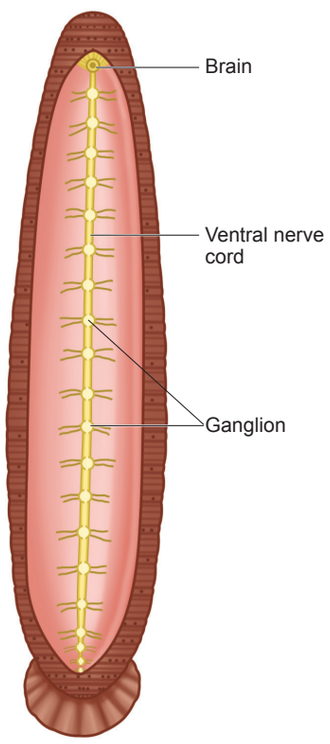


Figure 13.3 Nervous system of Leech

13.1.10 Excretory System

In leech, excretion takes place by segmentally arranged paired tubules called **nephridia**. There are 17 pairs of nephridia which open out by **nephridiopores** from 6th to 22nd segments.

13.1.11 Reproductive System

Leech is hermaphrodite because both the male and female reproductive organs are present in the same animal.

Male Reproductive System

There are eleven pairs of testes, one pair in each segment from 12 to 22 segments. They are in the form of spherical sacs called **testes sacs**. From each testis arises a short duct called **vas efferens**, which join with the vas deferens. The vas deferens becomes convoluted to form the **epididymis** or **sperm vesicle**, to store spermatozoa.

The epididymis leads to a short duct called ejaculatory duct. The **ejaculatory ducts** on both sides join to form the **genital atrium**. The atrium consists of two regions, the coiled prostate glands and the penial sac consisting of penis that opens through the male genital pore.

Female Reproductive System

It consists of ovaries, oviducts and vagina. There is a single pair of ovary in the 11th segment on the ventral side. Each ovary is a coiled ribbon-shaped structure.

The ova are budded off from the ovary. From each **ovary** runs a short oviduct. The **oviducts** of the two sides joins together, to form a common oviduct. The common oviduct opens into a pear-shaped **vagina** which lies mid-ventrally in the posterior part of the 11th segment.

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